RAMAIAH Institute of Technology

IEEE Computational Intelligence Society Bangalore Chapter

QRIT

QUANTUM WEEK '26

IBM Quantum

Q-HACK INDIA 2026

Team Name:

Team Lead Name:

Track:

Problem Statement title:



Q-SHIELD: End-to-End Quantum Security Platform

Discover, understand, protect and migrate quantum-vulnerable cryptography before Q-day

QUANTUM WEEK '26

IBM Quantum

Problem

Today's public-key cryptography has an expiry date

1

Shor breaks RSA & ECC

A large fault-tolerant quantum computer running Shor's algorithm derives private keys from public keys. RSA, ECDSA, ECDH and Diffie-Hellman all fall.

2

Harvest now, decrypt later

Attackers can record encrypted traffic and stolen data today, then decrypt it once quantum computers mature. Long-lived data is exposed already.

3

No inventory, no plan

Most organizations can't say where vulnerable cryptography lives in their code, configs and certificates, so they can't prioritize a migration that takes years.

The clock is set: NIST standardized ML-KEM, ML-DSA and SLH-DSA in 2024 (FIPS 203–205) and plans to deprecate quantum-vulnerable public-key algorithms after 2030 (NIST IR 8547, draft).

QUANTUM WEEK '26

IBM Quantum

Why It Matters

The deadline is set by your data, not by quantum computers

Who faces it

1. **Banks & payments**
   transaction records, card data, inter-bank keys
2. **Healthcare**
   medical records kept for decades
3. **Government & defense**
   classified and citizen data
4. **Any organization**
   with data that must stay secret 10+ years

Mosca's inequality: data is at risk when

$$X + Y > Z$$

X = years data must stay secret · Y = years to migrate · Z = years until Q-day

Example from our scan: QuantumBank payment-api



Result: RSA-protected financial records outlive their encryption, rated high exposure. Migration has to start now.

QUANTUM WEEK 26

IBM Quantum

Proposed Solution

One platform from discovery to migration

1 DISCOVER

Crypto scanner

Scans Python, Java, JS and Go code, SSH/JWT configs, X.509 certificates and committed keys into a Cryptographic Bill of Materials.

2 UNDERSTAND

Risk + quantum threat

Readiness score 0–100, harvest-now-decrypt-later exposure per data class, Shor and Grover simulated.

3 PROTECT

PQC gateway

Quantum-safe channel with ML-KEM-768 key exchange and ML-DSA-65 signatures, hybrid with X25519.

4 MIGRATE

Crypto-agility engine

A replacement for every finding, a 4-phase roadmap and the projected readiness after each phase.

5 AI SECURITY COPILOT Explains the risk, answers “what first?”, writes developer tickets and an executive report, grounded only in the scan.

QUANTUM WEEK '26

IBM Quantum

Why Quantum?

Quantum computing is both the threat and our method

The threat

Shor's algorithm

Broken

Exponential speedup for factoring and discrete logarithms: RSA and elliptic-curve keys can be derived from public keys.

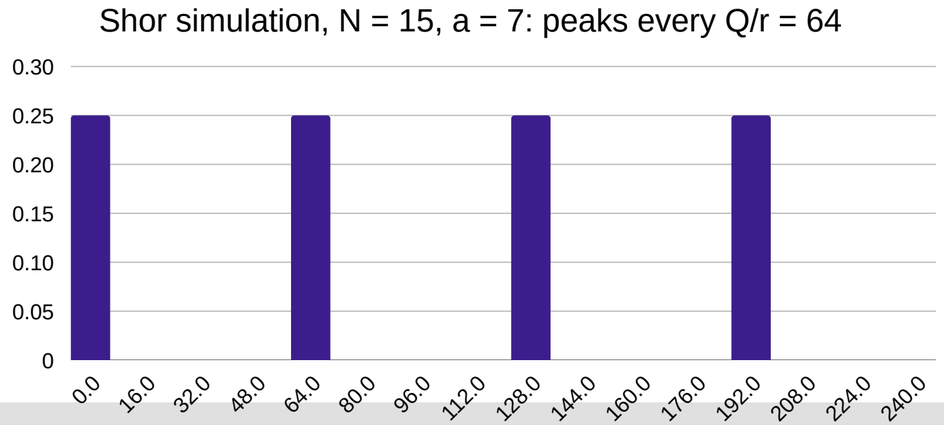
Grover's algorithm

Weakened

Quadratic speedup for brute-force search: AES-128 drops to about 64-bit strength; AES-256 keeps 128 bits.

Our method

Exact statevector simulation of Shor's period finding: modular exponentiation, inverse QFT, measurement, then continued fractions and gcd. Post-quantum ML-KEM/ML-DSA are built to resist exactly this adversary.



QUANTUM WEEK '26

IBM Quantum

Implementation and Architecture

One inventory feeds every module

Inputs

- Git repository
- ZIP upload
- Source code
- Configs
- Certificates & keys

FastAPI backend (Python)

- **Crypto scanner**
- **Risk engine**
- **HNDL analyzer**
- **Quantum simulator**
- **Migration engine**
- **PQC gateway**
- **AI copilot**

Cryptographic Bill of Materials (JSON schema) · SQLite

Outputs

- Dashboard (7 tabs)
- Quantum-safe chat
- Executive report
- Developer tickets
- CBOM export

quantum simulation post-quantum cryptography classical

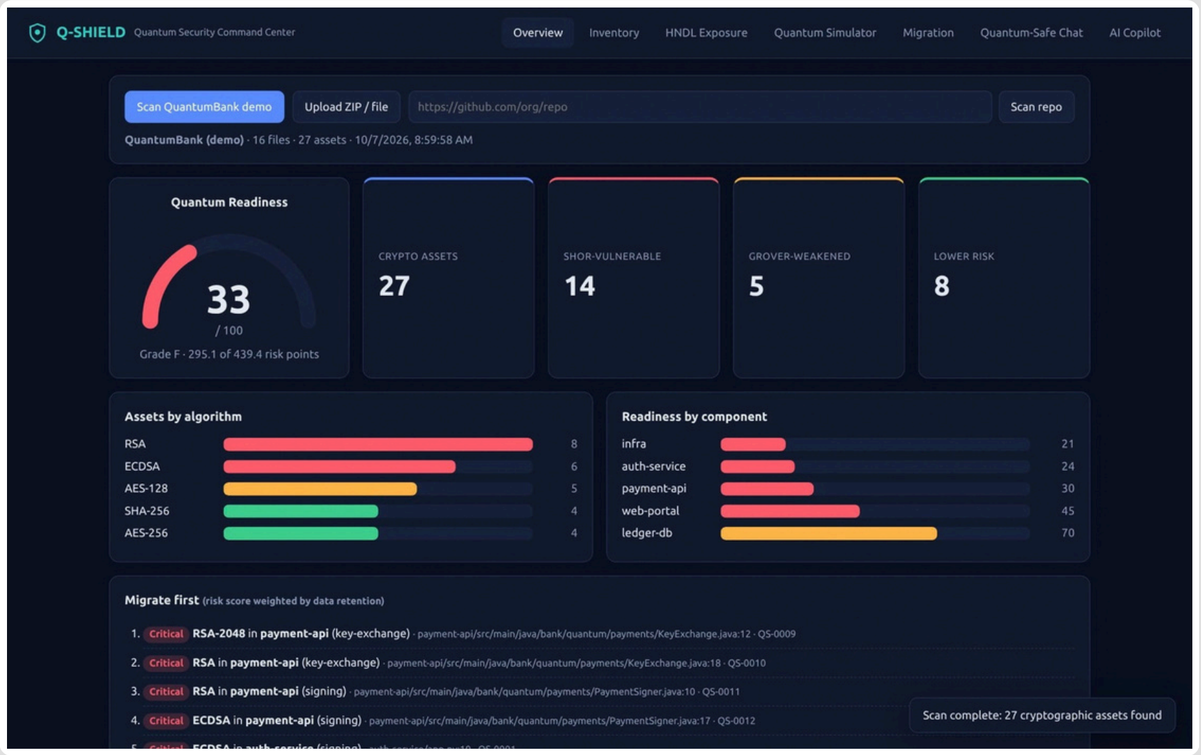
Stack: Python · FastAPI · pyca/cryptography (OpenSSL) for ML-KEM-768, ML-DSA-65, AES-256-GCM · NumPy statevector · Claude API · SQLite · plain JS + SVG · Docker

QUANTUM WEEK '26

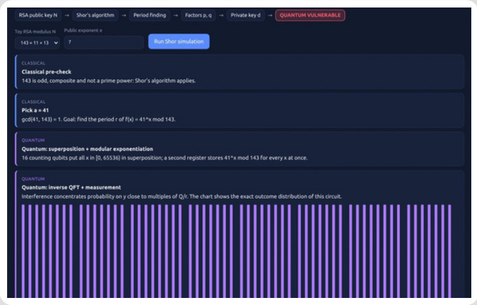
IBM Quantum

Minimal Prototype

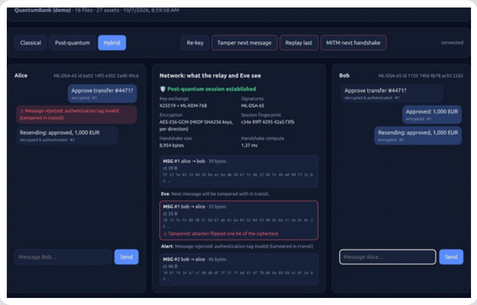
A working platform, not a mock-up



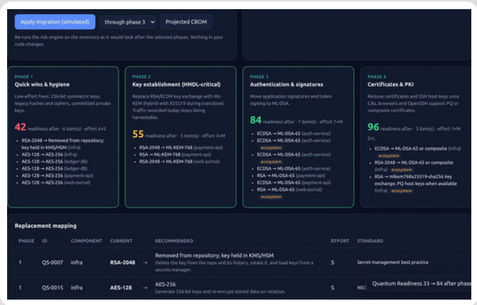
Overview: scan of QuantumBank, 27 assets, readiness 33/100



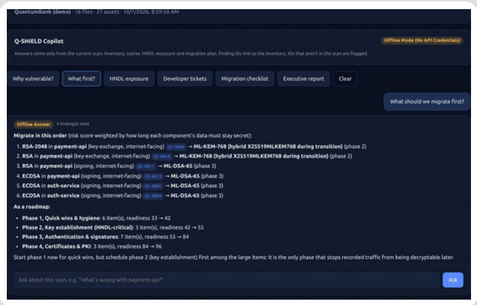
Shor simulation on 143



ML-KEM chat, tamper rejected



Roadmap: 33 → 84 → 96



Grounded AI copilot

QUANTUM WEEK 26

IBM Quantum

Results / Validation

Validated on a seeded bank, end to end

27 / 27

seeded crypto assets found in QuantumBank, 0 hits on commented-out code or docs

33 → 84

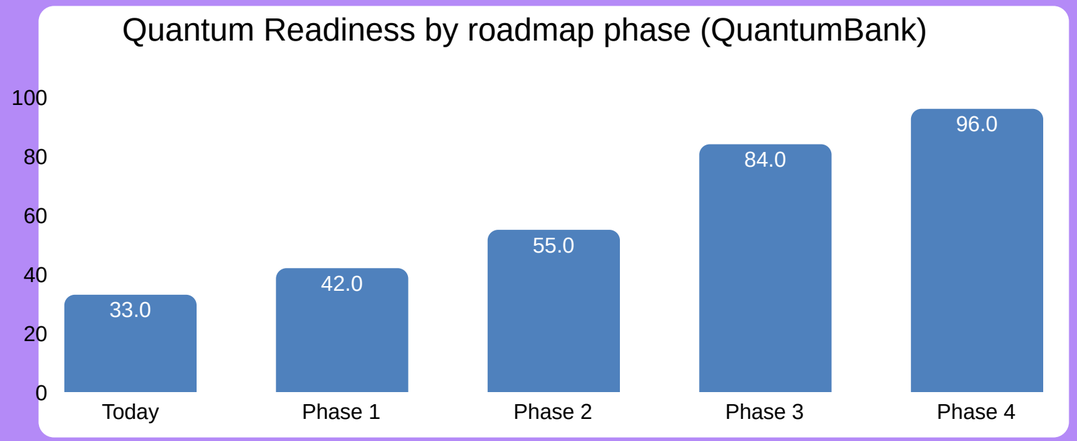
readiness after roadmap phases 1–3 (96 after phase 4), recomputed by the risk engine

8,954 B

hybrid ML-KEM-768 + ML-DSA-65 handshake, a few ms of compute

80

automated tests passing, including tamper, replay and MITM rejection



Also validated

- Shor simulation factors 15, 21, 143, 221 and 247; peaks match theory exactly
- ML-KEM/ML-DSA wire sizes match FIPS 203/204 (1184 B key, 1088 B ciphertext, 3309 B signature)
- Copilot cites only real finding IDs; invented IDs are flagged
- ZIP-slip archives and unsafe Git URLs rejected

QUANTUM WEEK '26

IBM Quantum

Roadmap

From demo app to real codebases

1 — 2 — 3 — 4

Deeper discovery

- AST and data-flow analysis
- Binaries and containers
- Live TLS/SSH endpoint scanning

Standards & CI

- CycloneDX CBOM export
- CI gate that blocks new vulnerable crypto
- GitHub app

Production PQC

- Hybrid TLS gateway (X25519MLKEM768)
- PQ / composite certificates
- KMS and HSM integration

Real validation

- Precision/recall on public repos
- Pilot with a partner organization
- Shor N = 15 on IBM Quantum hardware (Qiskit)

Round 2 goal: scan real repositories with measured accuracy, and run the period-finding circuit on real quantum hardware.



QUANTUM WEEK 26

IBM Quantum

References and Sources

Standards & research

- NIST FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA), Aug 2024. csrc.nist.gov/pubs/fips/203/final
- NIST IR 8547 (initial public draft), Transition to Post-Quantum Cryptography Standards, Nov 2024
- P. W. Shor, Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer, SIAM J. Computing 26(5), 1997
- L. K. Grover, A fast quantum mechanical algorithm for database search, STOC 1996
- M. Mosca, Cybersecurity in an era with quantum computers: will we be ready?, IEEE Security & Privacy 16(5), 2018
- C. Gidney, M. Ekerå, How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits, arXiv:1905.09749
- C. Gidney, How to factor 2048 bit RSA integers with less than a million noisy qubits, arXiv:2505.15917

Software & documentation

- pyca/cryptography: ML-KEM, ML-DSA, AES-GCM, HKDF. github.com/pyca/cryptography
- FastAPI, NumPy, Pydantic, Uvicorn
- Anthropic Claude API (copilot). platform.claude.com/docs
- CycloneDX CBOM specification. cyclonedx.org/capabilities/cbom
- Open Quantum Safe project (inspiration). openquantumsafe.org

Originality: Q-SHIELD is our own implementation for Q-Hack India 2026. Libraries are used as dependencies, not forked.